

Ashi Poorey

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EDUCATION

Bachelor of Science - Astrophysics

University of Illinois Urbana Champaign

Minors - Data Science and Computational Science & Engineering

May 2027

RESEARCH

Space Astronomy Summer Program intern - STScI

Jun 2026 - Aug 2026

Undergraduate Researcher [REU]

- Developed machine learning models to identify rare galaxies exhibiting broad Mg II emission in Sloan Digital Sky Survey spectra, targeting signatures of fading supermassive black holes
- Analyzed large spectroscopic datasets (~1.4 million sources) to isolate and validate a small sample of candidate objects, enabling study of black hole accretion duty cycles
- Designed scalable methods for deployment on future surveys (SDSS extensions, DESI, etc.) to discover and characterize transient active galactic nuclei in time-domain astronomy

UIS Stellar Evolution Research - Prof John Martin

Feb 2026 - present

Undergraduate Researcher

- Investigated photometric variability and evolutionary behavior of Warm Hypergiants in M31 and M33 near the Humphreys–Davidson limit and yellow void
- Analyzed time-series photometry using Lomb–Scargle periodograms and phase-folding techniques to identify periodic and quasi-periodic variability associated with stellar pulsations and mass loss
- Implementing red noise characterization methods to assess variability significance beyond false alarm probabilities, with ongoing work toward publication

UIUC Team Gaia - Prof Jiayin Dong

Aug 2025 - May 2026

Undergraduate Researcher

- Independently developed and implemented an exoplanet detection pipeline using ESA Gaia mission data, constructing Bayesian inference frameworks in PyTensor + PyMC
- Applied statistical methods to time-series astrometric data for exoplanet detection, with Bayesian statistics and probabilistic modeling
- Led research paper discussions and participated in hackathon-style coding sessions, strengthening skills in Python programming, scientific problem-solving, and teamwork

CONFERENCES, TALKS & PRESENTATIONS

1. **APS CU*IP 2026** - Gaia Pipeline for Inferring Planetary Companions using RUWE [*poster*]
2. **STScI SASP Symposium 2026** - Using Machine Learning to Discover Fading AGN [*presentation*]
3. **CAURS 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 [*poster*]

4. **UIUC URS 2026** - Gaia Pipeline for Inferring Planetary Companions using RUWE [*poster*]
5. **UIUC Astrofest 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 [*poster*]

PROGRAMS & TRAINING

STScI JWST Summer School – *High-Redshift Transients and Time Domain Astronomy* Aug 2025

- Selected to participate in a program combining lectures on high-z transient science with hands-on training in JWST observation planning, calibration pipelines, and data analysis.
- Gained practical experience in creating JWST observing programs and processing early-universe astrophysical data using advanced infrared techniques.

AWARDS

1. **Distinguished Undergraduate Researcher** - University of Illinois Urbana Champaign
2. **Celia Mathews Elliot Prize for Writing in Science and Engineering** - Department of Physics UIUC

OUTREACH, VOLUNTEERING & TEACHING

The Jeffries Academic Center - *Math & Physics Tutor* Jan 2025 - May 2025
Provided weekly academic support in Physics and Math to over five students

UIUC College of Liberal Arts & Sciences - *Student Ambassador* Sep 2025 - present
Representing the college at outreach and media events, and assisting prospective students with academics, campus life, and opportunities in LAS

UIUC LAS Lift Off 2025 - *Department of Astronomy booth* Aug 2025
Represented the Astronomy department, welcoming new students and answering questions about courses, research opportunities, and campus resources

UIUC New Student Welcome Week 2026 - *Orientation Leader* Aug 2026
Represented the student body of UIUC to implement orientation programming for new first-year and transfer students and family members, to help have a successful transition.

PROJECTS

Picture an Astronomer: Best Practices for Retaining Talent in Astrophysics | [Publication](#)
Co-authored research on diversity and retention in astrophysics, synthesizing literature and community insights into actionable recommendations

Radio Astronomy Capstone Project: S255IR NIRS3
Processed ALMA data using CASA and CARTA to produce continuum and C34S molecular line results, generating publication-quality figures and deriving astrophysical interpretations